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BACKLIGHT MODULE AND DISPLAY DEVICE

Abstract

A backlight module and a display device are disclosed. The light-guiding structure includes a first light-guiding surface opposite to the first light-emitting member and a second light-guiding surface opposite to the second light-emitting member, the first light-guiding surface has a specular reflective surface, and the second light-guiding surface has a diffuse reflective surface. When the backlight module is configured in the first state, the first light-emitting member emits lights, the second light-emitting member does not emit lights. When the backlight module is configured in the second state, at least the second light-emitting member emits lights.

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Background/Summary

TECHNICAL FIELD
[0001] The present application relates to the field of display technologies, and more particularly, to a backlight module and a display device.
BACKGROUND
[0002] Display devices are used in various aspects of people's daily life, and different applications require different viewing angles of the display devices. For example, when a user is in an open environment in which privacy needs to be kept, such as entering a withdrawal password, watching private information by public transportation, or negotiating a business, it is necessary to display with a narrow viewing angle to achieve anti-peeping so as to protect personal privacy. When a user is in an environment in which there is a need for sharing, such as viewing the display device together with others, a wide viewing angle is required to achieve the purpose of sharing.
[0003] Currently, an anti-peeping film, such as a liquid crystal anti-peeping film, is often added to the display device to achieve the anti-peeping in real time. However, an existing anti-peeping film is complex in structure which greatly increases the cost of the display device.
TECHNICAL SOLUTION
[0004] An embodiment of the present application provides a backlight module and a display device, which can realize anti-peeping in real time, simplify the structure of the backlight module, and reduce the costs.
[0005] An embodiment of the present application provides a backlight module including: [0006] a light-guiding board, where the light-guiding board includes a bottom surface away from a light-emitting side of the backlight module, and a first side surface and a second side surface connected to the bottom surface, and the bottom surface is provided with a plurality of light-guiding structures; [0007] a first light bar, where the first light bar includes a first light-emitting member disposed on a side of the first side surface, and the first light-emitting member has a light-emitting direction toward the first side surface and emits lights in the first direction; and [0008] a second light bar, where the second light bar includes a second light-emitting member disposed on a side of the second side surface, the second light-emitting member has a light-emitting direction toward the second side surface and emits lights in the second direction, and the first direction is different from the second direction; [0009] where each of the plurality of light-guiding structures includes a first light-guiding surface disposed opposite to the first light-emitting member and a second light-guiding surface disposed opposite to the second light-emitting member, the first light-guiding surface has a specular reflective surface, the second light-guiding surface has a diffuse reflective surface, and the backlight module is switchable between a first state and a second state; where in a case that the backlight module is configured to the first state, the first light-emitting member emits lights, and the second light-emitting member does not emit lights; in a case that the backlight module is configured to the second state, at least the second light-emitting member emits lights.
[0010] According to the above object of the present application, an embodiment of the present application further provides a display device including a display panel and a backlight module, where the display panel is provided on a light-emitting side of the backlight module; [0011] where the backlight module includes: [0012] a light-guiding board, where the light-guiding board includes a bottom surface away from a light-emitting side of the backlight module, and a first side surface and a second side surface connected to the bottom surface, and the bottom surface is provided with

a plurality of light-guiding structures; [0013] a first light bar, where the first light bar includes a first light-emitting member disposed on a side of the first side surface, and the first light-emitting member has a light-emitting direction toward the first side surface and emits lights in the first direction; and [0014] a second light bar, where the second light bar includes a second light-emitting member disposed on a side of the second side surface, the second light-emitting member has a light-emitting direction toward the second side surface and emits lights in the second direction, and the first direction is different from the second direction; [0015] where each of the plurality of light-guiding structures includes a first light-guiding surface disposed opposite to the first light-emitting member and a second light-guiding surface disposed opposite to the second light-emitting member, the first light-guiding surface has a specular reflective surface, the second light-guiding surface has a diffuse reflective surface, and the backlight module is switchable between a first state and a second state; where in a case that the backlight module is configured to the first state, the first light-emitting member emits lights, and the second light-emitting member does not emit lights; in a case that the backlight module is configured to the second state, at least the second light-emitting member emits lights.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. **1** is a schematic plan view of a backlight module according to an embodiment of the present application;

[0017] FIG. **2** is a schematic cross-sectional view of FIG. **1** taken along a line AA according to an embodiment of the present application;

[0018] FIG. **3** is a schematic structural diagram of a light-guiding structure according to an embodiment of the present application;

[0019] FIG. **4** is another schematic plan view of a backlight module according to an embodiment of the present application;

[0020] FIG. **5** is a schematic cross-sectional view of FIG. **4** taken along a line BB according to an embodiment of the present application;

[0021] FIG. **6** is a schematic cross-sectional view of FIG. **4** taken along a line CC according to an embodiment of the present application;

[0022] FIG. **7** is a schematic plan view of another backlight module according to an embodiment of the present application;

[0023] FIG. **8** is a schematic cross-sectional view of FIG. **7** taken along a line DD according to an embodiment of the present application;

[0024] FIG. **9** is a schematic cross-sectional view of FIG. **7** taken along a line EE according to an embodiment of the present application;

[0025] FIG. **10** is another plan view of a backlight module according to an embodiment of the present application;

[0026] FIG. **11** is a schematic cross-sectional view of FIG. **10** taken along a line FF according to an embodiment of the present application;

[0027] FIG. **12** is a schematic cross-sectional view of FIG. **10** taken along a line GG according to an embodiment of the present application; and

[0028] FIG. **13** is a schematic structural diagram of a display device according to an embodiment of the present application.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0029] The technical solution in the embodiments of the present application will be clearly and completely described with reference to the accompanying drawings. It should be understood that the described embodiments are only part of the embodiments of the present application, and not all embodiments. Based on the embodiments in the present application, all other embodiments obtained by a person skilled in the art without involving any inventive effort are within the scope of the present application.

[0030] The following disclosure provides many different embodiments or examples for implementing the different structures of the present application. In order to simplify the disclosure of the present application, components and arrangements are described in specific examples below. Of course, they are merely examples and are not intended to limit the application. In addition, the present application may repeat reference numerals and/or reference letters in various examples, such repetition is for the purpose of simplicity and clarity, without itself indicating a relationship between the various embodiments and/or arrangements discussed. In addition, the present application provides examples of various specific processes and materials, but one of ordinary skill in the art will recognize the application of other processes and/or the use of other materials.

[0031] An embodiment of the present application provides a backlight module. Referring to FIGS. **1** and **2**, the backlight module includes a light-guiding board **10**, a first light bar **21**, and a second light bar **22**.

[0032] The light-guiding board **10** includes a bottom surface **100** away from the light-exiting side of the backlight module, a first side surface **101** and a second side surface **102**. The first side surface **101** and the second side surface **102** are connected to the bottom surface **100**, and the bottom surface **100** is provided with a light-guiding structure **11**. The first light bar **21** includes a first light-emitting member **211** disposed on a side of the first side surface **101**, and the first light-emitting member **211** has a light-emitting direction toward the first side surface **101** and emits lights in the first direction X. The second light bar **22** includes a second light-emitting member **221** disposed on a side of the second side surface **102**, and the second light-emitting member **221** has a light-emitting direction toward the second side surface **102** and emits light in the second direction Y. The first direction X being different from the second direction Y.

[0033] The light-guiding structure **11** includes a first light-guiding surface **111** disposed opposite to the first light-emitting member **211**, and a second light-guiding surface **112** disposed opposite to the second light-emitting member **221**. The first light-guiding surface **111** has a specular reflective surface, the second light-guiding surface **112** has a diffuse reflective surface, and the backlight module is switched between a first state and a second state. When the backlight module is configured to the first state, the first light-emitting member **211** emits lights, and the second light-emitting member **221** does not emit lights. When the backlight module is configured to the second state, at least the second light-emitting member **221** emits lights.

[0034] In an implementation and application process, the embodiments of the present application provide the light-guiding structure **11** of the light-guiding board **10** in the backlight module, and provide the first light-guiding surface **111** of the light-guiding structure **11** opposite to the first light-emitting member **211** as a specular reflective surface and the second light-guiding surface **112** of the light-guiding structure **11** opposite to the second light-emitting member **221** as a diffuse reflective surface, so that the first light bar **21** emits lights with a small viewing angle, and the second light bar **22** emits lights with a large viewing angle. When the backlight module is configured in the first state, the first light-emitting member **211** emits lights, and the second light-emitting member **221** does not emit lights. As such, when the lights emitted from the first light-emitting member **211** pass through the first light-guiding surface **111** with the specular reflective surface, a small viewing angle is provided to achieve the anti-peeping effect, and the first state of the backlight module is the anti-peeping state. When the backlight module is configured in the second state, at least the second light-emitting member **221** emits lights. As such, when the lights emitted from the second light-emitting member **221** pass through the second light-guiding surface **112** with the diffuse reflective surface, a large viewing angle is provided to achieve the sharing effect, and the second state of the backlight module is in the sharing state. Thus, the anti-peeping function in real time is realized, and the structure of the backlight module is simplified, which

reduces the cost.
[0035] In an embodiment of the present application, an angle between the first light-guiding surface and the first direction is less than an angle between the second light-guiding surface and the second direction.
[0036] In an embodiment of the present application, an angle between the first light-guiding surface and the first direction is greater than or equal to 15° and less than or equal to 30°.
[0037] In an embodiment of the present application, the first light-guiding surface is planar and the second light-guiding surface is an arc surface.
[0038] In an embodiment of the present application, the second light-guiding surface projects toward a side away from the first light-guiding surface.
[0039] In an embodiment of the present application, the light-guiding board further includes a third side surface connected to the bottom surface, and the backlight module further includes a third light bar. The third light bar includes a third light-emitting member disposed on a side of the third side surface, the third light-emitting member has a light-emitting direction toward the third side surface and emits lights in a third direction, and the third direction is different from the first direction and the second direction.
[0040] The light-guiding structure includes a third light-guiding surface disposed opposite to the third light-emitting member, the third light-guiding surface has a specular reflective surface, and the third light-emitting member emits lights at least when the backlight module is configured in the first state.
[0041] In an embodiment of the present application, the third direction is opposite to the first direction.
[0042] In an embodiment of the present application, the light-guiding board further includes a fourth side surface connected to the bottom surface, the backlight module further includes a fourth light bar, and the fourth light bar includes a fourth light-emitting member disposed on a side of the fourth side surface. The fourth light-emitting member has a light-emitting direction toward the fourth side surface and emits lights in a fourth direction, and the fourth direction is different from the first direction and the second direction.
[0043] The light-guiding structure includes a fourth light-guiding surface disposed opposite to the fourth light-emitting member, and the fourth light-guiding surface has a diffuse reflective surface. The fourth light-emitting member does not emit lights when the backlight module is configured in the first state, and the fourth light-emitting member emits lights when the backlight module is configured in the second state.
[0044] In an embodiment of the present application, the fourth direction intersects the second direction, or the fourth direction is opposite to the second direction.
[0045] In an embodiment of the present application, the first light bar includes a first circuit board, the first light-emitting member is disposed on the first circuit board, the first circuit board is located on a side of the light-guiding board provided with the bottom surface, an end of the first circuit board is connected to the first light-emitting member, and the other end of the first circuit board overlaps the light-guiding board in a thickness direction of the light-guiding board.
[0046] The second light strip includes a second circuit board, the second light-emitting member is arranged on the second circuit board, the second circuit board is located on a side of the light-guiding board provided with the bottom surface, an end of the second circuit board is connected with the second light-emitting member, and the other end of the second circuit board overlaps the light-guiding board in the thickness direction of the light-guiding board.
[0047] In an embodiment of the present application, the first circuit board is provided with a first reflective layer on a surface opposite the light-guiding board, and the second circuit board is provided with a second reflective layer on a surface opposite the light-guiding board.
[0048] Specifically, with continued reference to FIGS. **1** and **2**, the backlight module provided in the present embodiment includes a light-guiding board **10**, and the light-guiding board **10** includes

a bottom surface **100** away from the light-emitting side of the backlight module, a first side surface **101**, and a second side surface **102**. The first side surface **101** and the second side surface **102** are connected to the bottom surface **100**. For example, when the light-guiding board **10** is a cuboid, the light-guiding board **10** has four side surfaces **102**, as shown in FIG. **1**.

[0049] The backlight module further includes a plurality of light bars disposed on the side surfaces of the light-guiding board **10**. Specifically, the backlight module includes a first light bar **21** and a second light bar **22** disposed on the peripheral side of the light-guiding board **10**. The first light bar **21** includes a first light-emitting member **211** disposed on a side of the first side surface **101** of the light-guiding board **10**, the first light-emitting member **211** emits lights toward the first side surface **101**, and the first light-emitting member **211** emits lights in the first direction X. The second light bar **22** includes a second light-emitting member **221** disposed on a side of the second side surface **102** of the light-guiding board **10**, the second light-emitting member **221** emits lights toward the second side surface **102**, and the second light-emitting member **221** emits lights in the second direction Y.

[0050] Here, the first direction X and the second direction Y are different, that is, the first light-emitting member **211** and the second light-emitting member **221** are located on different sides of the light-guiding board **10**, for example, the first direction X may intersect with the second direction Y, or the first direction X may be opposite to the second direction Y.

[0051] In an embodiment, when the first direction X intersects with the second direction Y, the first direction X may be perpendicular to the second direction Y.

[0052] It should be noted that the first light bar **21** may include a plurality of first light-emitting members **211**, and the first light-emitting members **211** may be LED light members provided on the first light bar **21**. The plurality of first light-emitting members **211** may be arranged in a direction parallel to the first side surface **101** of the light-guiding board **10**. Similarly, the second light bar **22** may include a plurality of second light-emitting members **221**, and the second light-emitting members **221** may be LED light members provided on the second light bar **22**. The plurality of second light-emitting members **221** may be arranged in a direction parallel to the second side **102** of the light-guiding board **10**.

[0053] In an embodiment, the first light bar **21** includes a first circuit board **212**, and the first light-emitting member **211** is located on the first side **101** of the light-guiding board **10**, the first circuit board **212** is located on a side of the light-guiding board **10** close to the bottom surface **100**, and the first light-emitting member **211** is disposed on the first circuit board **212**. An end of the first circuit board **212** is connected to the first light-emitting member **211**, and the other end of the first circuit board **212** overlaps with the light-guiding board **10** in the thickness direction of the light-guiding board **10**. Further, the second light bar **22** includes a second circuit board **222**, and the second light-emitting member **221** is located on the second side **102** of the light-guiding board **10**, the second circuit board **222** is located on a side of the light-guiding board **10** close to the bottom surface **100**, and the second light-emitting member **221** is disposed on the second circuit board **222**. An end of the second circuit board **222** is connected to the second light-emitting member **221**, and the other end of the second circuit board **222** overlaps with the light-guiding board **10** in the thickness direction of the light-guiding board **10**.

[0054] In addition, in order to avoid that the first circuit board **212** and the second circuit board **222** overlapped with the light-guiding board **10** to affect the lights emitted from the backlight module, an embodiment of the present application provides a first reflective layer **213** on the side of the first circuit board **212** adjacent to the light-guiding board **10**, and a second reflective layer **223** on the side of the second circuit board **222** adjacent to the light-guiding board **10**. The first reflective layer **213** is provided on the surface of the first circuit board **212** facing the light-guiding board **10**, and the second reflective layer **223** is provided on the surface of the second circuit board **222** facing the light-guiding board **10**. Further, the lights emitted from the first light-emitting member **211** and the second light-emitting member **221** to the bottom surface **100** of the light-guiding board **10** can be reflected to improve the light-emitting efficiency of the backlight module.

[0055] In an embodiment, the backlight module further includes a reflective sheet provided on the bottom surface **100** of the light-guiding board **10**, a diffuser sheet and a prism sheet provided on the light-emitting side of the light-guiding board **10** close to the backlight module, to further improve the light-emitting effect of the backlight module.

[0056] In the embodiment of the present application, the bottom surface **100** of the light-guiding board **10** is provided with the light-guiding structure **11** to adjust light emitted from the first light-emitting member **211** and the second light-emitting member **221** so that lights are emitted in a predetermined direction.

[0057] The light-guiding structure **11** includes a first light-guiding surface **111** disposed opposite to the first light-emitting member **211**, and a second light-guiding surface **112** disposed opposite to the second light-emitting member **221**. The first light-guiding surface **111** has a specular reflective surface such that the lights emitted from the first light-emitting member **211** are collimated lights. The second light-guiding surface **112** has a diffuse reflective surface such that the lights emitted from the second light-emitting member **221** are divergent lights.

[0058] It should be understood that the first light-guiding surface **111** is a surface of the light-guiding structure **11** facing the first light-emitting member **211**, and the second light-guiding surface **112** is a surface of the light-guiding structure **11** facing the second light-emitting member **221**.

[0059] Further, the backlight module includes the first state and the second state. When the backlight module is configured in the first state, the first light-emitting member **211** emits lights and the second light-emitting member **221** does not emit lights. As such, the backlight module emits collimated lights in the first state. When the backlight module is configured in the second state, at least the second light-emitting member **221** emits lights. As such, the backlight module emits divergent lights in the second state. In the present application, the light-guiding structure **11** of the light-guiding board **10** in the backlight module is designed to provide the first light-guiding surface **111** of the light-guiding structure **11** opposite to the first light-emitting member **211** as a specular reflective surface and the second light-guiding surface **112** of the light-guiding structure **11** opposite to the second light-emitting member **221** as a diffuse reflective surface, so that the lights emitted from the first light bar **21** has a small viewing angle and the lights emitted from the second light bar **22** has a large viewing angle. When the backlight module is configured in the first state, the first light-emitting member **211** emits lights, and the second light-emitting member **221** does not emit lights; and when the backlight module is configured in the second state, at least the second light-emitting member **221** emits lights. As such, the first state of the backlight module is an anti-peeping state, and the second state is a sharing state. The light-emitting angle of the backlight module in the first state ranges less than the light-emitting angle of the backlight module in the second state. In order to realize the anti-peeping function in real time, the structure of the backlight module is simplified, and the cost is reduced.

[0060] In an embodiment, as shown in FIG. **3**, the angle a between the first light-guiding surface **111** and the first direction X is less than the angle b between the second light-guiding surface **112** and the second direction Y, so that the lights emitted from the first light-emitting member **211** can be emitted in the forward direction after being guided by the first light-guiding surface **111**, and preferably, when the backlight module is configured in the first state, the light emitted from the backlight module is emitted in the forward direction.

[0061] In an embodiment, the angle between the first light-guiding surface **111** and the first direction X is greater than or equal to 15°, and less than or equal to 30°.

[0062] It should be noted that the bottom surface **100** of the light-guiding board **10** is provided with a plurality of light-guiding structures **11**, and the plurality of light-guiding structures **11** may be arranged in an array of a plurality of rows and a plurality of columns. Alternatively, in other embodiments of the present application, the plurality of light-guiding structures **11** may be arranged differently according to the luminance distribution of different areas, and this is not limited herein.

[0063] In an embodiment of the present application, with reference to FIGS. **1** and **2**, in the present embodiment, the first direction X is opposite to the second direction Y, that is, the first light-emitting member **211** and the second light-emitting member **221** are located on opposite sides of the light-guiding board **10**, respectively, and are oppositely illuminated.

[0064] The first light-guiding surface **111** and the second light-guiding surface **112** of the light-guiding structure **11** are respectively located on opposite sides of the light-guiding structure **11** so as to guide lights for the first light-emitting member **211** and the second light-emitting member **221**, respectively. When the backlight module is configured in the first state, the first light-emitting member **211** may be turned on, and the second light-emitting member **221** may be turned off, so that the backlight module has a small viewing angle, thereby achieving the anti-peeping function. When the backlight module is configured in the second state, the second light-emitting member **221** may be turned on, the first light-emitting member **211** may be turned off, or the second light-emitting member **221** and the first light-emitting member **211** may be turned on, so as to realize a large viewing angle of the backlight module and the sharing function.

[0065] In another embodiment of the present application, referring to FIG. **4**, FIG. **5**, and FIG. **6**, in the present embodiment, the light-guiding board **10** further includes a third side surface **103** connected to the bottom surface **100**, the backlight module further includes a third light bar **23**, and the third light bar **23** includes a third light-emitting member **231** disposed on a side of the third side surface **103**. The third light-emitting member **231** has a light-emitting direction toward the third side surface **103**, and the third light-emitting member **231** emits lights in a third direction M, where the third direction M is different from the first direction X and the second direction Y.

[0066] The third light bar **23** further includes a third circuit board **232**, the third light-emitting member **231** is provided on the third circuit board **232**, and the third circuit board **232** is located on the side of the light-guiding board **10** provided with the bottom surface **100**. An end of the third circuit board **232** is connected to the third light-emitting member **231**, and the other end of the third circuit board **232** overlaps with the light-guiding board **10** in the thickness direction of the light-guiding board **10**. In order to avoid that the part of the third circuit board **232** that overlapped with the light-guiding board **10** affects the light emission of the backlight module, an embodiment of the present application provides a third reflective layer **233** on a side of the third circuit board **232** adjacent to the light-guiding board **10**. The third reflective layer **233** is provided on a surface of the third circuit board **232** facing the light-guiding board **10**. Further, the lights emitted from the first light-emitting member **211**, the second light-emitting member **221**, and the third light-emitting member **231** to the bottom surface **100** of the light-guiding board **10** can be reflected to improve the light-emitting efficiency of the backlight module.

[0067] Alternatively, the third direction M may intersect with the first direction X, or the third direction M may be opposite to the first direction X. In the present embodiment, the third direction M is opposite to the first direction X, that is, the first side **101** and the third side **103** are opposite sides of the light-guiding board **10**, and the second side **102** is connected between the first side **101** and the third side **103**.

[0068] The third light-emitting member **231** and the first light-emitting member **211** are located on opposite sides of the light-guiding board **10**, and lights emitted from the third light-emitting member **231** and the first light-emitting member **211** are toward each other. The second direction Y intersects with the first direction X and intersects with the third direction M.

[0069] In the present embodiment, the light-guiding structure **11** includes a third light-guiding surface **113** disposed opposite to the third light-emitting member **231**, the third light-guiding surface **113** is a surface of the light-guiding structure **11** facing the third light-emitting member **231**, and the third light-guiding surface **113** has a specular reflective surface. As such, the lights emitted from the third light-emitting member **231** can be emitted toward the positive direction after being guided by the third light-guiding surface **113**, and preferably, when the backlight module is configured in the first state, the light-emitting direction of the backlight module is in the positive direction.

[0070] It should be noted that, since the light-guiding structure **11** can be patterned to form the bottom surface **100** of the light-guiding board **10** by using a laser or a cutter, when the first light-guiding surface **111** and the third light-guiding surface **113** are identical, the first light-guiding surface **111** and the third light-guiding surface **113** are located on opposite sides of the light-guiding structure **11** to reduce the processing difficulty of the light-guiding structure **11** and to simplify the manufacturing process of the light-guiding board **10**.

[0071] In an embodiment, the depth of the light-guiding structure **11** may be greater than or equal to 5 microns, and less than or equal to 15 microns, for example, may be 5 microns, 6 microns, 7 microns, 8 microns, 9 microns, 10 microns, 11 microns, 12 microns, 13 microns, 14 microns, or 15 microns. The bottom width of the light-emitting side of the light-guiding structure **11** away from the backlight module may be greater than or equal to 30 microns, and less than or equal to 70 microns, for example, may be 30 microns, 35 microns, 40 microns, 45 microns, 50 microns, 55 microns, 60 microns, 65 microns, or 70 microns, and the above values are for illustration only and are not limited thereto.

[0072] In an embodiment, the angle between the third light-guiding surface **113** and the third direction M is greater than or equal to 15° and less than or equal to 30°. For example, the angle may be 15°, 20°, 25°, or 30°, and the above values are for illustration only and are not limited thereto.

[0073] Further, the second direction Y may be perpendicular to the first direction X and the third direction M.

[0074] It should be understood that the light-guiding structure **11** may have four faces; three of the four faces are the first light-guiding face **111**, the second light-guiding face **112**, and the third light-guiding face **113**, respectively; and the remaining face of the four faces is disposed opposite to the second light-guiding face **112**, and has a diffuse reflective face, to reduce the manufacturing process difficulty of the light-guiding structure **11**.

[0075] According to the embodiment of the present application, the light-guiding structure **11** of the light-guiding board **10** in the backlight module is designed to provide the first light-guiding surface **111** of the light-guiding structure **11** opposite to the first light-emitting member **211** as a specular reflective surface and the second light-guiding surface **112** of the light-guiding structure **11** opposite to the second light-emitting member **221** as a diffuse reflective surface, so that the first light bar **21** emits the lights with a small viewing angle and the second light bar **22** emits the lights with large viewing angle. When the backlight module is configured in the first state, the first light-emitting member **211** and the third light-emitting member **231** emit lights, and the second light-emitting member **221** does not emit lights. When the backlight module is configured in a second state, the second light-emitting member **221** emits lights, or both the first light-emitting member **211**, the second light-emitting member **221** and the third light-emitting member **231** emit lights, so that the first state of the backlight module is in the anti-peeping state, and the second state is the sharing state. Thus, the anti-peeping function in real time is realized, the structure of the backlight module is simplified, and the cost is reduced.

[0076] In another embodiment of the present application, referring to FIG. **7**, FIG. **8**, and FIG. **9**, the present embodiment differs from the embodiment shown in FIG. **5** in that the light-guiding board **10** further includes a fourth side surface **104** connected to the bottom surface **100**, the backlight module further includes a fourth light bar **24**, the fourth light bar **24** includes a fourth light-emitting member **241** disposed on a side of the fourth side surface **104**, the fourth light-emitting member **241** has a light-emitting direction toward the fourth side surface **104** and emits lights along the fourth direction N, and the fourth direction N is different from the first direction X and the second direction Y.

[0077] The fourth direction N is opposite to the second direction Y, that is, the fourth light-emitting member **241** and the second light-emitting member **221** are respectively located on opposite sides of the light-guiding board **10**, and the fourth light-emitting member **241** and the second light-emitting member **221** emit lights toward each other.
[0078] The light-guiding structure **11** includes a fourth light-guiding surface **114** disposed opposite to the fourth light-emitting member **241**. The fourth light-guiding surface **114** is a surface of the light-guiding structure **11** facing the fourth light-emitting member **241**. The fourth light-guiding surface **114** has a diffuse reflective surface. When the backlight module is configured in the first state, the fourth light-emitting member **241** does not emit lights. When the backlight module is configured in the second state, the fourth light-emitting member **241** emits lights.
[0079] The fourth light bar **24** further includes a fourth circuit board **242** on which the fourth light-emitting member **241** is provided, and the fourth circuit board **242** is located on a side of the light-guiding board **10** provided with the bottom surface **100**. An end of the fourth circuit board **242** is connected to the fourth light-emitting member **241**, and the other end of the fourth circuit board **242** overlaps with the light-guiding board **10** in the thickness direction of the light-guiding board **10**. In order to avoid that the part of the fourth circuit board **242** overlapped with the light-guiding board **10** affects the light emission of the backlight module, the fourth reflective layer **243** is provided on the side of the fourth circuit board **242** adjacent to the light-guiding board **10**, and the fourth reflective layer **243** is provided on a surface of the fourth circuit board **242** facing the light-guiding board **10**. Further, the lights emitted from the first light-emitting member **211**, the second light-emitting member **221**, the third light-emitting member **231**, and the fourth light-emitting member **241** to the bottom surface **100** of the light-guiding board **10** can be reflected to improve the light-emitting efficiency of the backlight module.
[0080] According to the embodiment of the present application, the light-guiding structure **11** of the light-guiding board **10** in the backlight module is designed to provide the first light-guiding surface **111** of the light-guiding structure **11** opposite to the first light-emitting member **211** as a specular reflective surface and the second light-guiding surface **112** of the light-guiding structure **11** opposite to the second light-emitting member **221** as a diffuse reflective surface, so that the first light bar **21** emits the lights with a small viewing angle and the second light bar **22** emits the lights with large viewing angle. When the backlight module is configured in the first state, the first light-emitting member **211** and the third light-emitting member **231** emit lights, and the second light-emitting member **221** does not emit lights. When the backlight module is configured in a second state, the second light-emitting member **221** and the fourth light-emitting member **241** emit lights, or all the first light-emitting member **211**, the second light-emitting member **221**, the third light-emitting member **231** and the fourth light-emitting member **241** emit lights, so that the first state of the backlight module is in the anti-peeping state, and the second state is the sharing state. Thus, the anti-peeping function in real time is realized, the structure of the backlight module is simplified, and the cost is reduced.
[0081] In another embodiment of the present application, referring to FIG. **10**, FIG. **11** and FIG. **12**, the present embodiment differs from the embodiment shown in FIG. **1** in that the light-guiding board **10** further includes a fourth side **104** connected to the bottom surface **100** and a fifth side **105** connected to the bottom surface **100**, and the backlight module further includes a fourth light bar **24** and a fifth light bar **25**.
[0082] Specifically, the fourth light bar **24** includes a fourth light-emitting member **241** disposed on a side of the fourth side surface **104**, the fourth light-emitting member **241** has a light-emitting direction toward the fourth side surface **104** and emits lights in the fourth direction N, and the fourth direction N is different from both the first direction X and the second direction Y, where the fourth direction N intersects the second direction Y.
[0083] The light-guiding structure **11** includes a fourth light-guiding surface **114** disposed opposite to the fourth light-emitting member **241**. The fourth light-guiding surface **114** has a diffuse

reflective surface. When the backlight module is configured in the first state, the fourth light-emitting member **241** does not emit lights. When the backlight module is configured in the second state, the fourth light-emitting member **241** emits lights.

[0084] The fifth light bar **25** includes a fifth light-emitting member **251** disposed on a side of the fifth side surface **105**, the fifth light-emitting member **251** has a light-emitting direction toward the fifth side surface **105** and emits lights in the fifth direction P, and the fifth direction P is different from the first direction X and the second direction Y. Here, the fifth direction P is opposite to the fourth direction N, that is, the fourth light-emitting member **241** and the fifth light-emitting member **251** are provided on opposite sides of the light-guiding board **10**, respectively, and the fourth light-emitting member **241** and the fifth light-emitting member **251** emit lights toward each other.

[0085] The light-guiding structure **11** includes a fifth light-guiding surface **115** disposed opposite to the fifth light-emitting member **251**. The fifth light-guiding surface **115** is a surface of the light-guiding structure **11** facing the fifth light-emitting member **251**. The fifth light-guiding surface **115** has a diffuse reflective surface. When the backlight module is configured in the first state, the fifth light-emitting member **251** does not emit lights. When the backlight module is configured in the second state, the fifth light-emitting member **251** emits lights.

[0086] In the present embodiment, the first side **101** and the second side **102** are opposite sides of the light-guiding board **10**, and the fourth side **104** and the fifth side **105** are opposite sides of the light-guiding board **10**.

[0087] Further, the fourth light bar **24** further includes a fourth circuit board **242** on which the fourth light-emitting member **241** is provided, and the fourth circuit board **242** is located on a side of the light-guiding board **10** provided with the bottom surface **100**, an end of the fourth circuit board **242** is connected to the fourth light-emitting member **241**, and the other end of the fourth circuit board **242** overlaps with the light-guiding board **10** in the thickness direction of the light-guiding board **10**. In order to avoid that the part of the fourth circuit board **242** overlapped with the light-guiding board **10** affects the light emission of the backlight module, the fourth reflective layer **243** is provided on the side of the fourth circuit board **242** adjacent to the light-guiding board **10**, and the fourth reflective layer **243** is provided on the surface of the fourth circuit board **242** facing the light-guiding board **10**. Further, the lights emitted from the first light-emitting member **211**, the second light-emitting member **221**, the fourth light-emitting member **241**, and the fifth light-emitting member **251** to the bottom surface **100** of the light-guiding board **10** can be reflected to improve the light-emitting efficiency of the backlight module.

[0088] The fifth light bar **25** further includes a fifth circuit board **252** on which the fifth light-emitting member **251** is provided, and the fifth circuit board **252** is located on a side of the light-guiding board **10** provided with the bottom surface **100**. An end of the fifth circuit board **252** is connected to the fifth light-emitting member **251**, and the other end of the fifth circuit board **252** overlaps with the light-guiding board **10** in the thickness direction of the light-guiding board **10**. In order to avoid that the part of the fifth circuit board **252** overlapped with the light-guiding board **10** affects the light emission of the backlight module, an embodiment of the present application provides a fifth reflective layer **253** on a side of the fifth circuit board **252** adjacent to the light-guiding board **10**, and the fifth reflective layer **253** is provided on a surface of the fifth circuit board **252** facing the light-guiding board **10**. Further, the lights emitted from the first light-emitting member **211**, the second light-emitting member **221**, the fourth light-emitting member **241**, and the fifth light-emitting member **251** to the bottom surface **100** of the light-guiding board **10** can be reflected to improve the light output efficiency of the backlight module.

[0089] According to the embodiment of the present application, the light-guiding structure **11** of the light-guiding board **10** in the backlight module is designed to provide the first light-guiding surface **111** of the light-guiding structure **11** opposite to the first light-emitting member **211** as a specular reflective surface and the second light-guiding surface **112** of the light-guiding structure **11** opposite to the second light-emitting member **221** as a diffuse reflective surface, so that the first light bar **21** emits the lights with a small viewing angle and the second light bar **22** emits the lights with large viewing angle. When the backlight module is configured in the first state, the first light-emitting member **211**, the second light-emitting member **221**, the fourth light-emitting member **241** and the fifth light-emitting member **251** do not emit lights. When the backlight module is configured in a second state, the second light-emitting member **221**, the fourth light-emitting member **241**, and the fifth light-emitting member **251** emit lights, or all the first light-emitting member **211**, the second light-emitting member **221**, the fourth light-emitting member **241** and the fifth light-emitting member **251** emit lights, so that the first state of the backlight module is in the anti-peeping state, and the second state is the sharing state. Thus, the anti-peeping function in real time is realized, the structure of the backlight module is simplified, and the cost is reduced.

[0090] It should be understood that the first light-guiding surface **111** and the third light-guiding surface **113** in the above-described embodiment are smooth surfaces and can perform specular reflection, that the second light-guiding surface **112**, the fourth light-guiding surface **114**, and the fifth light-guiding surface **115** are rough surfaces in circular arc shapes and can perform diffuse reflection, and that the second light-guiding surface **112** protrudes toward a side away from the first light-guiding surface **111**, and that the fourth light-guiding surface **114** and the fifth light-guiding surface **115** protrude toward a side away from the center of the light-guiding structure **11**, so that light emitted from the second light-emitting member **221**, the fourth light-emitting member **241**, and the fifth light-emitting member **251** can be better dispersed and dispersed to a larger range of the viewing angle, thereby improving the viewing angle range and the light-emitting effect of the backlight module in the second state.

[0091] It should be noted that, in the embodiments of the present application, the light-guiding board **10** is described in terms of a top view shape as a quadrangle, and the light-guiding board **10** may be provided with other polygons, so that each light-emitting member is located on a side surface of the light-guiding board **10** and emits light toward the direction close to the light-guiding board **10**. The light-guiding structure **11** in the light-guiding board **10** forms a light-guiding surface with respect to a side surface of each light-emitting member, and at least one specular reflective surface and one diffuse reflecting surface need to be formed. The specular reflective surface may enable the lights emitted from the light-emitting member to be emitted in a collimated manner so as to realize the light emission at the small viewing angle, and the diffuse reflective surface may enable the lights emitted from the light-emitting member to be emitted in a divergent manner so as to realize the light emission at the large viewing angle, thereby enabling the backlight module to switch between the anti-peeping state and the sharing state.

[0092] In an embodiment of the present application, the light-guiding structure **11** is uniformly distributed in the light-guiding board **10**. In other embodiments of the present application, the backlight module includes a first region and a second region, and since the brightness of the original first region is less than that of the second region, the distribution density of the light-guiding structure **11** in the first region is set greater than that of the light-guiding structure **11** in the second region in the embodiments of the present application. As such, the average brightness of the first region is improved to be equal to the average brightness of the second region, thereby improving the problem of uneven brightness distribution in the backlight module. For example, the first region may be an area around the backlight module, the second region may be an area in the middle of the backlight module. The area around the backlight module may be reduced in brightness due to loss of lights shielded by the frame. The embodiment of the present application may improve brightness uniformity of the backlight module. Further, the first region and the second region may be other regions in the backlight module in which the brightness distribution is not uniform, and are not limited herein.

[0093] In addition, an embodiment of the present application further provides a display device. Referring to FIG. **1** and FIG. **13**, the display device includes a display panel **40** and a backlight module **30** as described in the above-described embodiment, and the display panel **40** is disposed on the light-emitting side of the backlight module **30**.

[0094] When the backlight module **30** is configured in the first state, the light emission with small viewing angle can be realized, the display device can realize the anti-peeping mode, and when the backlight module **30** is configured in the second state, the light emission with large viewing angle can be realized, the display device can realize the sharing mode. Therefore, in the embodiment of the present application, by controlling the turn-on and turn-off of the light-emitting member on the side surface of the light-guiding board **10**, the anti-peeping mode and the sharing mode of the display device can be switchable, and no additional structure such as an anti-peeping film is needed, thereby simplifying the process and reducing the cost.

[0095] In the above-mentioned embodiments, the description of each embodiment has its own emphasis, and parts not described in detail in a certain embodiment may be referred to the related description of other embodiments.

[0096] The present application has been described in detail with reference to a backlight module and a display device according to an embodiment of the present invention. The principles and embodiments of the present application are described herein using specific examples. The description of the above embodiments is merely provided to help understand the technical solution and the core idea of the present application. It should be understood by those of ordinary skill in the art that modifications may still be made to the technical solutions described in the foregoing embodiments, or equivalents may be made to some of the technical features therein. These modifications or equivalents do not depart the essence of the corresponding technical solutions from the scope of the technical solutions of the embodiments of the present application.

Claims

1. A backlight module, comprising: a light-guiding board, wherein the light-guiding board comprises a bottom surface away from a light-emitting side of the backlight module, and a first side surface and a second side surface connected to the bottom surface, and the bottom surface is provided with a plurality of light-guiding structures; a first light bar, wherein the first light bar comprises a first light-emitting member disposed on a side of the first side surface, and the first light-emitting member has a light-emitting direction toward the first side surface and emits lights in the first direction; and a second light bar, wherein the second light bar comprises a second light-emitting member disposed on a side of the second side surface, the second light-emitting member has a light-emitting direction toward the second side surface and emits lights in the second direction, and the first direction is different from the second direction; wherein each of the plurality of light-guiding structures includes a first light-guiding surface disposed opposite to the first light-emitting member and a second light-guiding surface disposed opposite to the second light-emitting member, the first light-guiding surface has a specular reflective surface, the second light-guiding surface has a diffuse reflective surface, and the backlight module is switchable between a first state and a second state; wherein in a case that the backlight module is configured to the first state, the first light-emitting member emits lights, and the second light-emitting member does not emit lights; in a case that the backlight module is configured to the second state, at least the second light-emitting member emits lights.

2. The backlight module of claim 1, wherein an angle included between the first light-guiding surface and the first direction is less than an angle included between the second light-guiding surface and the second direction.

3. The backlight module of claim 1, wherein the angle included between the first light-guiding surface and the first direction is greater than or equal to 15° and less than or equal to 30°.

4. The backlight module of claim 1, wherein the first light-guiding surface is a planar surface and the second light-guiding surface is an arc surface.

5. The backlight module of claim 4, wherein the second light-guiding surface protrudes toward a side away from the first light-guiding surface.
6. The backlight module of claim 1, wherein the light-guiding board further comprises a third side surface connected to the bottom surface, the backlight module further comprises a third light bar, the third light bar comprises a third light-emitting member disposed on a side of the third side surface, the third light-emitting member has a light-emitting direction toward the third side surface and emits lights in a third direction, and the third direction is different from the first direction and the second direction; the light-guiding structure includes a third light-guiding surface disposed opposite to the third light-emitting member, the third light-guiding surface has a specular reflective surface, and the third light-emitting member emits lights at least when the backlight module is configured in the first state.
7. The backlight module of claim 6, wherein the third direction is opposite to the first direction.
8. The backlight module of claim 1, wherein the light-guiding board further comprises a fourth side surface connected to the bottom surface, the backlight module further comprises a fourth light bar, the fourth light bar comprises a fourth light-emitting member disposed on a side of the fourth side surface, the fourth light-emitting member has a light-emitting direction toward the fourth side surface and emits lights in a fourth direction, and the fourth direction is different from the first direction and the second direction; the light-guiding structure includes a fourth light-guiding surface disposed opposite to the fourth light-emitting member, the fourth light-guiding surface has a diffuse reflective surface, the fourth light-emitting member does not emit lights when the backlight module is configured in the first state, and the fourth light-emitting member emits lights when the backlight module is configured in the second state.
9. The backlight module of claim 8, wherein the fourth direction intersects with the second direction, or the fourth direction is opposite to the second direction.
10. The backlight module of claim 1, wherein the first light bar includes a first circuit board, the first light-emitting member is disposed on the first circuit board, the first circuit board is located on a side of the light-guiding board provided with the bottom surface, an end of the first circuit board is connected to the first light-emitting member, and another end of the first circuit board overlaps with the light-guiding board in a thickness direction of the light-guiding board; the second light strip comprises a second circuit board, the second light-emitting member is disposed on the second circuit board, the second circuit board is located on a side of the light-guiding board provided with the bottom surface, an end of the second circuit board is connected with the second light-emitting member, and another end of the second circuit board overlaps with the light-guiding board in the thickness direction of the light-guiding board.
11. The backlight module of claim 10, wherein a surface of the first circuit board facing the light-guiding board has a first reflective layer, and a surface of the second circuit board facing the light-guiding board has a second reflective layer.
12. A display device, comprising a display panel and a backlight module, wherein the display panel is disposed on a light-emitting side of the backlight module; wherein the backlight module comprises: a light-guiding board, wherein the light-guiding board comprises a bottom surface away from a light-emitting side of the backlight module, and a first side surface and a second side surface connected to the bottom surface, and the bottom surface is provided with a plurality of light-guiding structures; a first light bar, wherein the first light bar comprises a first light-emitting member disposed on a side of the first side surface, and the first light-emitting member has a light-emitting direction toward the first side surface and emits lights in the first direction; and a second light bar, wherein the second light bar comprises a second light-emitting member disposed on a side of the second side surface, the second light-emitting member has a light-emitting direction toward the second side surface and emits lights in the second direction, and the first direction is different from the second direction; wherein each of the plurality of light-guiding structures includes a first light-guiding surface disposed opposite to the first light-emitting member and a

second light-guiding surface disposed opposite to the second light-emitting member, the first light-guiding surface has a specular reflective surface, the second light-guiding surface has a diffuse reflective surface, and the backlight module is switchable between a first state and a second state; wherein in a case that the backlight module is configured to the first state, the first light-emitting member emits lights, and the second light-emitting member does not emit lights; in a case that the backlight module is configured to the second state, at least the second light-emitting member emits lights.

13. The display device of claim 12, wherein an angle included between the first light-guiding surface and the first direction is less than an angle included between the second light-guiding surface and the second direction.

14. The display device of claim 12, wherein the angle included between the first light-guiding surface and the first direction is greater than or equal to 15° and less than or equal to 30°.

15. The display device of claim 12, wherein the first light-guiding surface is a planar surface and the second light-guiding surface is an arc surface.

16. The display device of claim 15, wherein the second light-guiding surface protrudes toward a side away from the first light-guiding surface.

17. The display device of claim 12, wherein the light-guiding board further comprises a third side surface connected to the bottom surface, the backlight module further comprises a third light bar, the third light bar comprises a third light-emitting member disposed on a side of the third side surface, the third light-emitting member has a light-emitting direction toward the third side surface and emits lights in a third direction, and the third direction is different from the first direction and the second direction; the light-guiding structure includes a third light-guiding surface disposed opposite to the third light-emitting member, the third light-guiding surface has a specular reflective surface, and the third light-emitting member emits lights at least when the backlight module is configured in the first state.

18. The display device of claim 17, wherein the third direction is opposite to the first direction.

19. The display device of claim 12, wherein the light-guiding board further comprises a fourth side surface connected to the bottom surface, the backlight module further comprises a fourth light bar, the fourth light bar comprises a fourth light-emitting member disposed on a side of the fourth side surface, the fourth light-emitting member has a light-emitting direction toward the fourth side surface and emits lights in a fourth direction, and the fourth direction is different from the first direction and the second direction; the light-guiding structure includes a fourth light-guiding surface disposed opposite to the fourth light-emitting member, the fourth light-guiding surface has a diffuse reflective surface, the fourth light-emitting member does not emit lights when the backlight module is configured in the first state, and the fourth light-emitting member emits lights when the backlight module is configured in the second state.

20. The display device of claim 19, wherein the fourth direction intersects with the second direction, or the fourth direction is opposite to the second direction.